Mapping the Regener-Pfotzer Maximum: A Global Collaboration in Cosmic Ray Astrophysics

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Background

What are Cosmic Rays?

- Discovered in 1912 through a high-altitude balloon experiment done by Victor Francis Hess [1].
- Highly energetic, ionized nuclei made of protons, alpha particles, and heavy nuclei [2].
- Generated from astrophysical phenomena, such as solar activity and supernovae [2].
- When cosmic rays interact with out atmosphere, they create a shower of particles.
  - a source of ionizing radiation [3].
- This research attempts to measure the ionizing radiation flux as a function of altitude.
  - The altitude at which the ionizing radiation flux reaches a maximum is referred to as the *Regener-Pfotzer Maximum*.

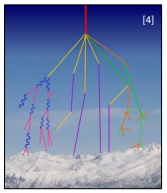


Figure 1: Illustration of Particle Showers resulting from Cosmic Ray interactions with the Earth's atmosphere

What is HERA?

- The Earth and Sun's magnetic field can modulate the cosmic ray flux [3].
- The High-Altitude Engineering Research for Astrophysics (HERA) Collaborative Project aims to study this modulation by conducting coordinated high-altitude balloon launches at various latitudes.

Methods

- Coordinated balloon flights from PA, CA, TX, and SC reached ~30 km around local noon, carrying similar radiation sensing payloads such as scintillators (Cosmic Watch and mDetect) and Geiger counters.
- Other payloads were included such as SPOT, APRS and data loggers to record UTC-stamped position, pressure, temperature, magnetic field, and count summaries every 30 seconds.
- An APRS interactive map auto-populates pop-ups for each launch site with key details. Coded with Python(Folium) + leaflet.js HTML.
- Post-flight CSV logs were time-aligned, binned into minute-wide CPM, and plotted to highlight the Regener-Pfotzer peak across various latitudes.

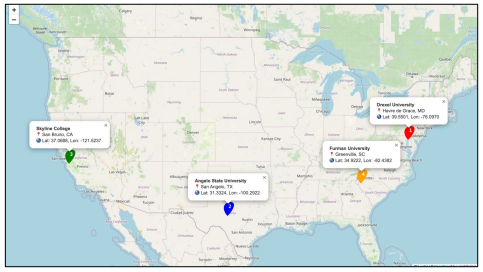
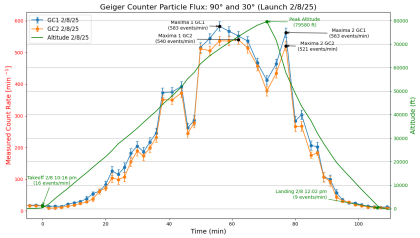


Figure 2: Map of the universities participating in the HERA collaboration and their respective launch sites for the May 3rd, 2025 coordinated high-altitude balloon launch.

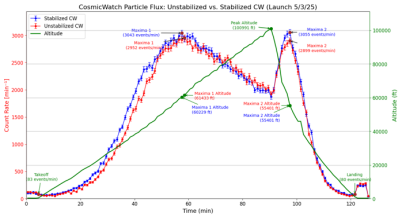
Results

Solar-Alignment Influence on Count Rate

The Geiger Counter oriented at 90° to the sun consistently recorded higher particle flux than the unittilted at 30°.



Solar-aligned, vertically stabilized Cosmic Watch recorded higher particle flux than the unstabilized detector.

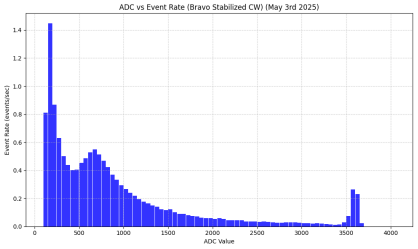


Collectively, the Geiger Counter and Cosmic Watch results confirm that cosmic particles arriving at an angle traverse a greater atmospheric path, reducing the measured flux compared to vertical incidence.



ADC Readout – Event Frequency

ADC amplitude is proportional to the energy deposited in the scintillator, and this plot depicts the ADC values recorded by the stabilized Cosmic Watch to characterize its particle interaction response.



The CosmicWatch ADC distribution exhibits a dominant peak at ≈ 700 ADC units, consistent with minimum-ionizing particle energy deposition, accompanied by a subtler peak near 3500 ADC units, indicative of higher-energy secondary interactions.

Discussion

- **Solar-Alignment Influence on Count Rate:**
  - Geiger Counters: The Geiger counter exhibits a systematic alignment error of ~3 %, yet positioned at 90° it still records approximately 15 % higher flux than at 30°, underscoring the need to correct for angular biases when interpreting count-rate variations.
  - CosmicWatch Detectors: The stabilized Cosmic Watch registers ~3 % greater peak flux at the first maximum (3043 vs 2952 events/min) and ~5 % at the second (3055 vs 2899 events/min) compared to the unstabilized unit, highlighting the impact of vertical stabilization on measured rates. A brief dip in the stabilized CW's count rate between 60 min and 90 mi, around peak altitude, likely reflects high-altitude orientation drift; reviewing magnetometer and accelerometer logs and applying in-flight sensor recalibration or filtering that interval can help address this.
- **ADC Readout - Event Frequency:**
  - The pronounced low-energy peak at the left of the ADC spectrum stems from electronic noise and carries no physical significance for particle detection; the peaks at ≈700 ADC (minimum-ionizing events) and ≈3500 ADC (higher-energy secondaries) are therefore the primary features informing our event-frequency analysis.

Future Work

- Expand HERA worldwide.
- Upgrade detectors with multi-layer scintillators & higher-resolution SiPMs.
- Add new environmental sensors & deploy satellite-based IoT for advanced telemetry.
- Enhance analysis with Seaborn (a Python library) visualizations and ML-based peak detection, including ADC value plotting from mDetect for deeper analysis.





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